

What is climate change?

ECON-4210

Economics of Climate Change

Class 2

Prof. Casey Wichman

Today's class

1. Questions about course policies / syllabus?
2. Attendance
3. Crash course in (basic) climate science
4. For next week...

Climate change is not new.

Published: Aug 14, 1912.

COAL CONSUMPTION AFFECT- ING CLIMATE.

The furnaces of the world are now burning about 2,000,000,000 tons of coal a year. When this is burned, uniting with oxygen, it adds about 7,000,000,000 tons of carbon dioxide to the atmosphere yearly. This tends to make the air a more effective blanket for the earth and to raise its temperature. The effect may be considerable in a few centuries.

Published Aug. 14, 1912.

THE RODNEY AND OTAMATEA TIMES AND WAITEMATA AND KAIPARA GAZETTE

THE
LONDON, EDINBURGH, AND DUBLIN
PHILOSOPHICAL MAGAZINE
AND
JOURNAL OF SCIENCE.

[FIFTH SERIES.]

APRIL 1896.

XXXI. *On the Influence of Carbonic Acid in the Air upon the Temperature of the Ground.* By Prof. SVANTE ARRHENIUS*.

I. *Introduction: Observations of Langley on Atmospheric Absorption.*

A GREAT deal has been written on the influence of the absorption of the atmosphere upon the climate. Tyndall† in particular has pointed out the enormous importance of this question. To him it was chiefly the diurnal and annual variations of the temperature that were lessened by this circumstance. Another side of the question, that has long attracted the attention of physicists, is this: Is the mean temperature of the ground in any way influenced by the presence of heat-absorbing gases in the atmosphere? Fourier‡ maintained that the atmosphere acts like the glass of a hot-house, because it lets through the light rays of the sun but retains the dark rays from the ground. This idea was elaborated by Pouillet§; and Langley was by some of his researches led to the view, that “the temperature of the earth under direct sunshine, even though our atmosphere were present as now, would probably fall to -200° C., if that atmosphere did not possess the quality of selective

* Extract from a paper presented to the Royal Swedish Academy of Sciences, 11th December, 1895. Communicated by the Author.

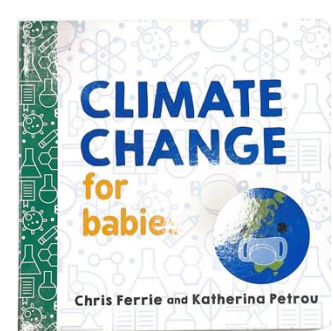
† ‘Heat a Mode of Motion,’ 2nd ed. p. 405 (Lond., 1865).

‡ *Mém. de l’Ac. R. d. Sci. de l’Inst. de France*, t. vii. 1827.

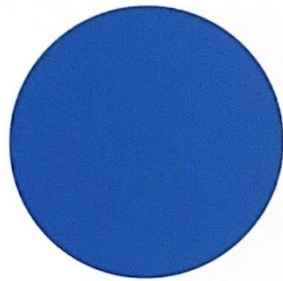
§ *Comptes rendus*, t. vii. p. 41 (1838).

Basic climate science

*Disclaimer: I am not a climate or physical scientist



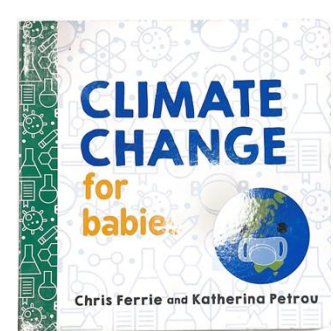
An extremely simplified (but also useful and fun) intro to climate change



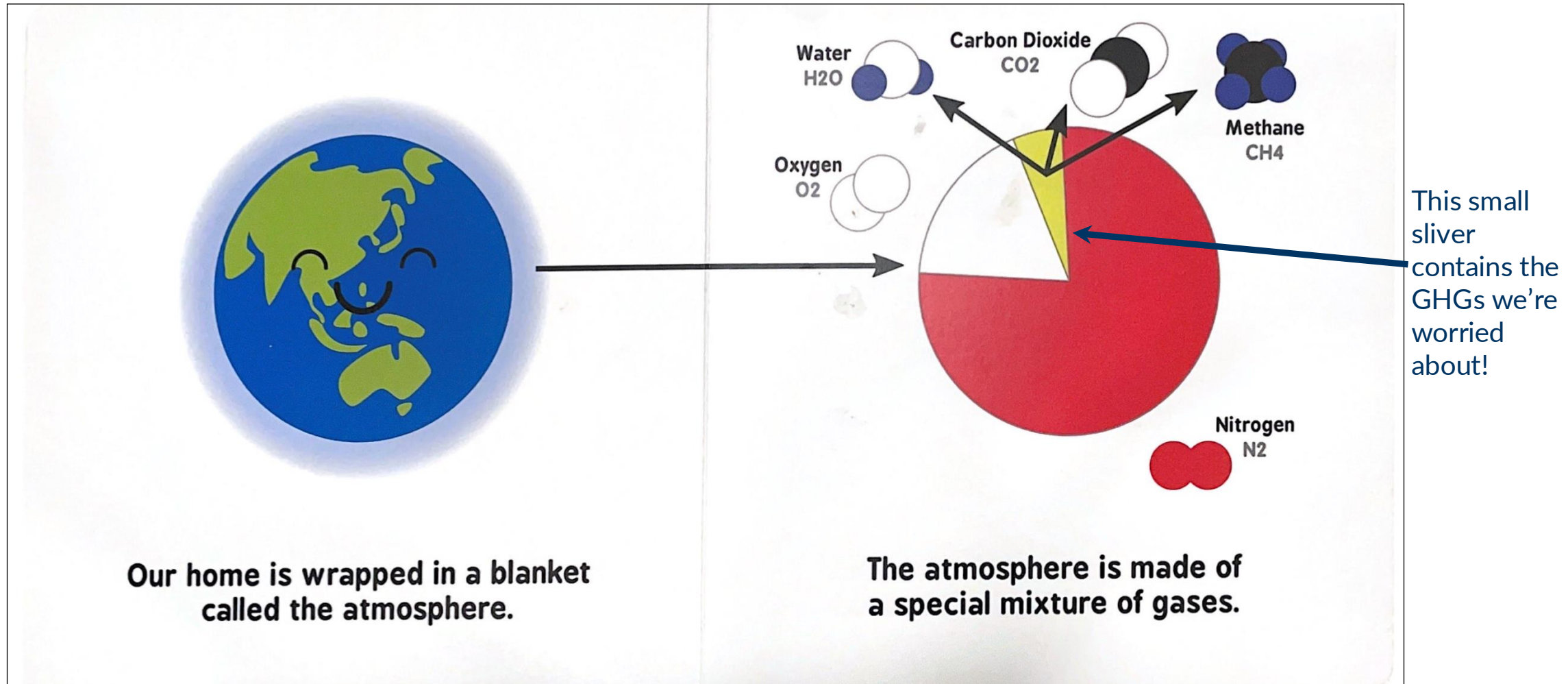
This is a ball.



It is the shape of Earth, our home.



An extremely simplified (but also useful and fun) intro to climate change



An extremely simplified (but also useful and fun) intro to climate change

**CLIMATE
CHANGE**

for
babies



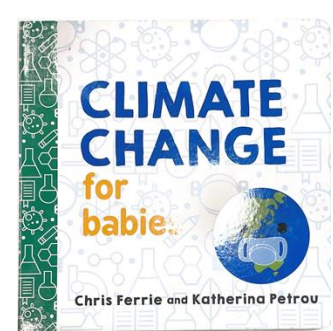
Chris Ferrie and Katherina Petrou



**It protects the Earth from
the cold of outer space**



by trapping just the right amount of heat.



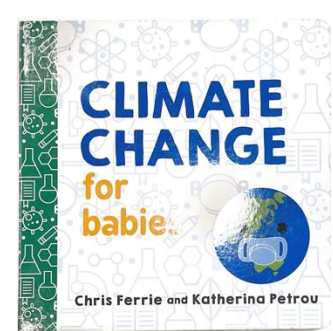
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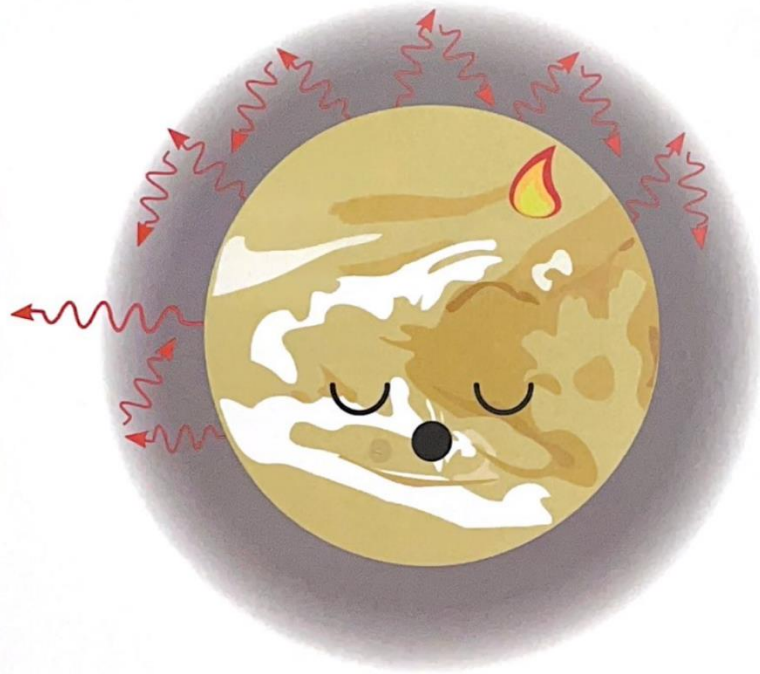
Earth's blanket is just right thanks to trees, animals, and oceans.



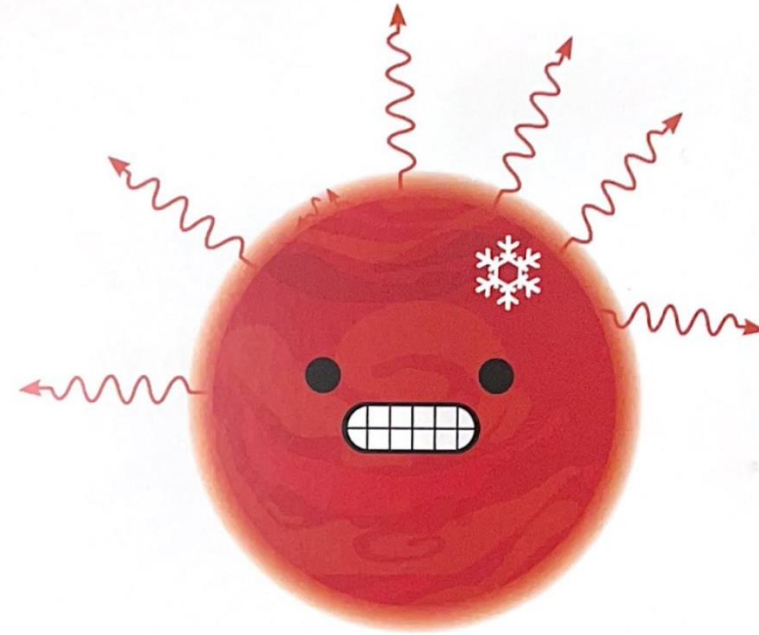
They help keep the atmosphere in balance.



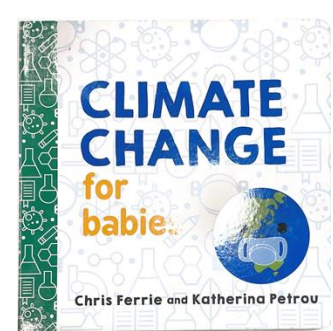
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The planet Venus has a much thicker blanket and is too hot.



The planet Mars has a thin blanket and is too cold.



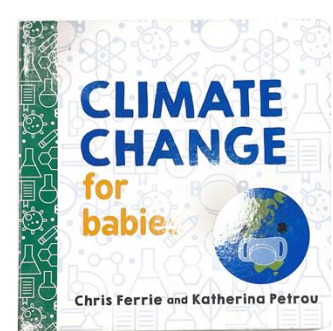
An *extremely* simplified (but also useful and fun) intro to climate change



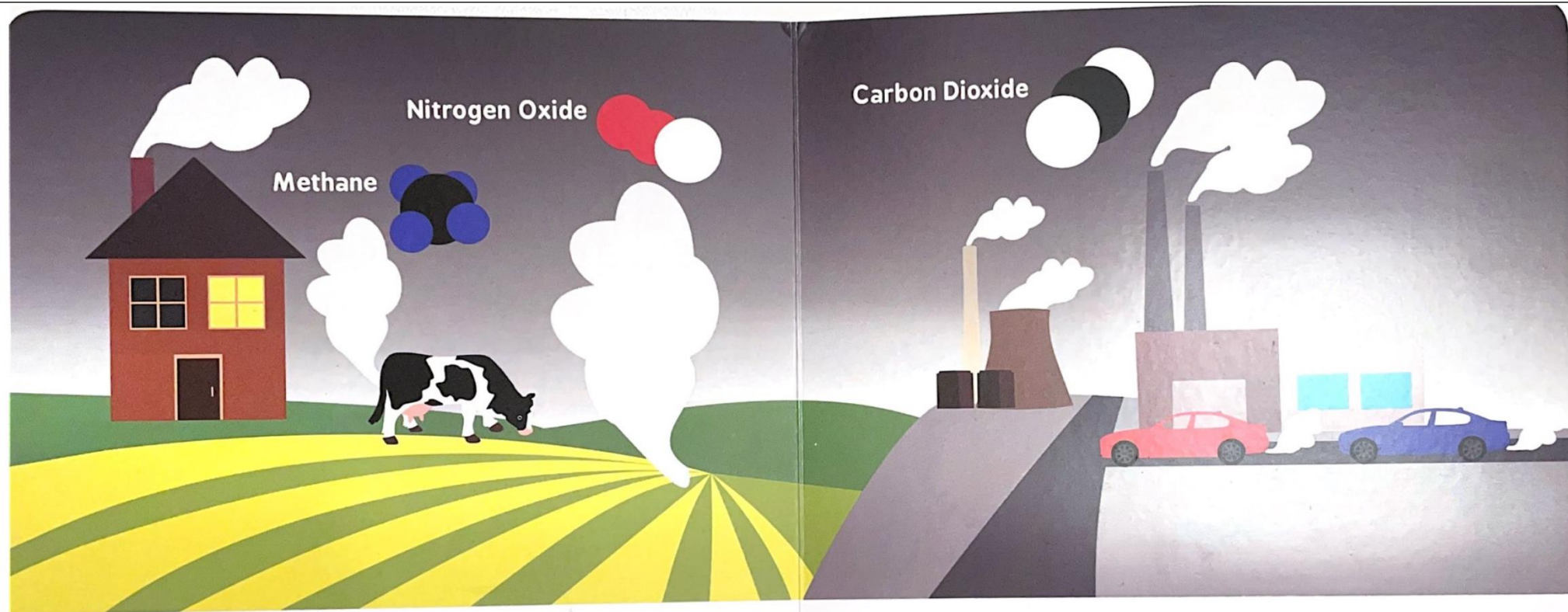
**But... Uh-oh! People on Earth
are changing the blanket!**



**Industry, heating, coal power,
transportation, livestock, and fertilizers...**

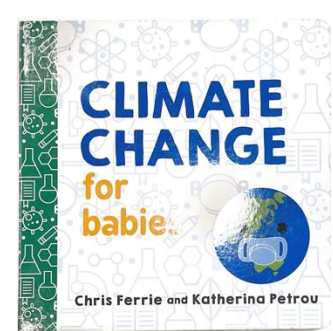


An *extremely* simplified (but also useful and fun) intro to climate change



all send greenhouse gases

into the atmosphere.



An extremely simplified (but also useful and fun) intro to climate change

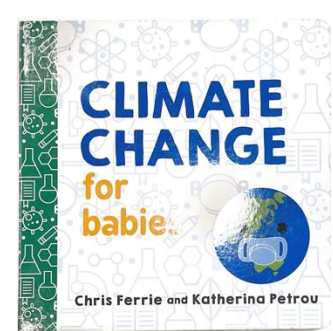


This is making the blanket heavier and thicker.

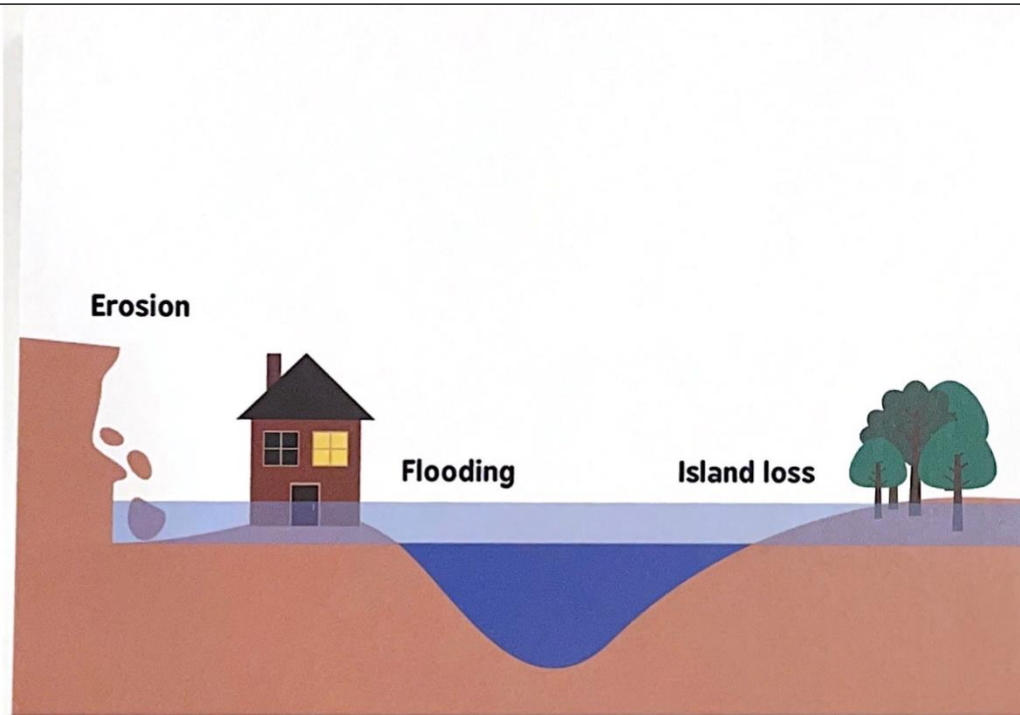


A thicker blanket traps heat, making Earth feel unwell.

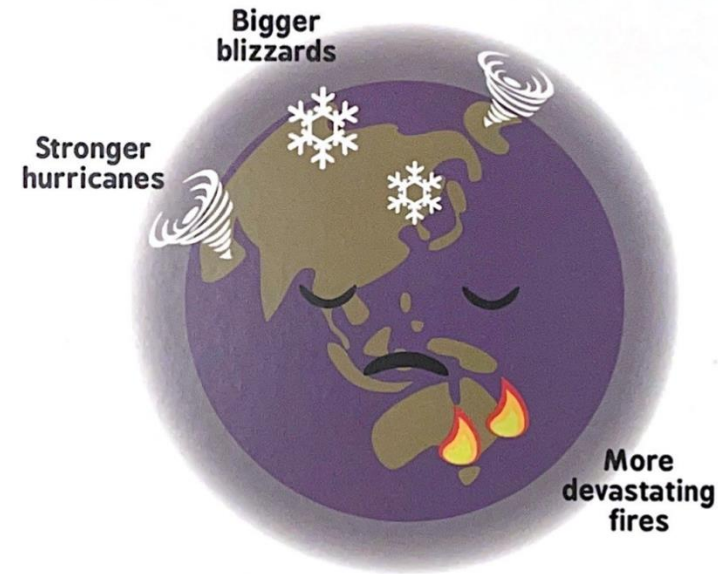
Well... not exactly,
but let's roll with
the metaphor...



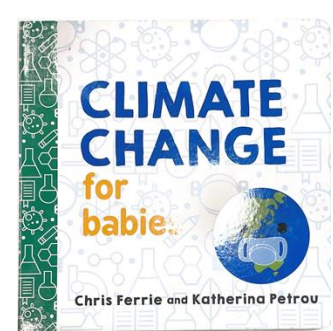
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**Warmer temperatures cause
the sea level to rise**

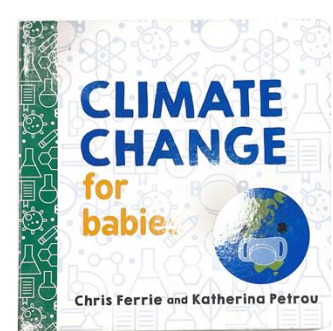


and cause more extreme weather events.

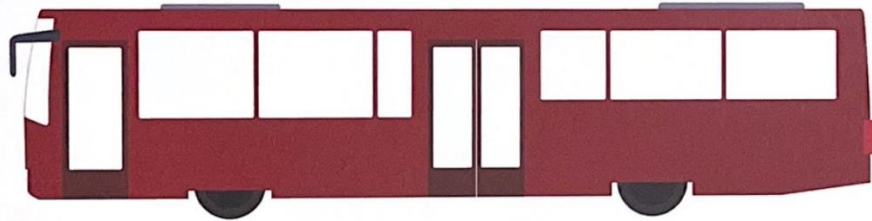


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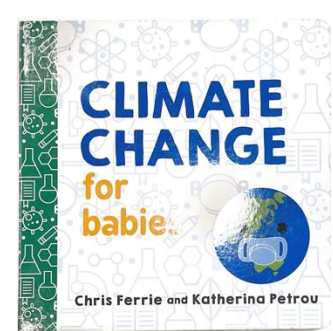
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Fewer cars and factories mean less greenhouse gases in the atmosphere.



We can replace coal and oil with solar and wind power.



An extremely simplified (but also useful and fun) intro to climate change



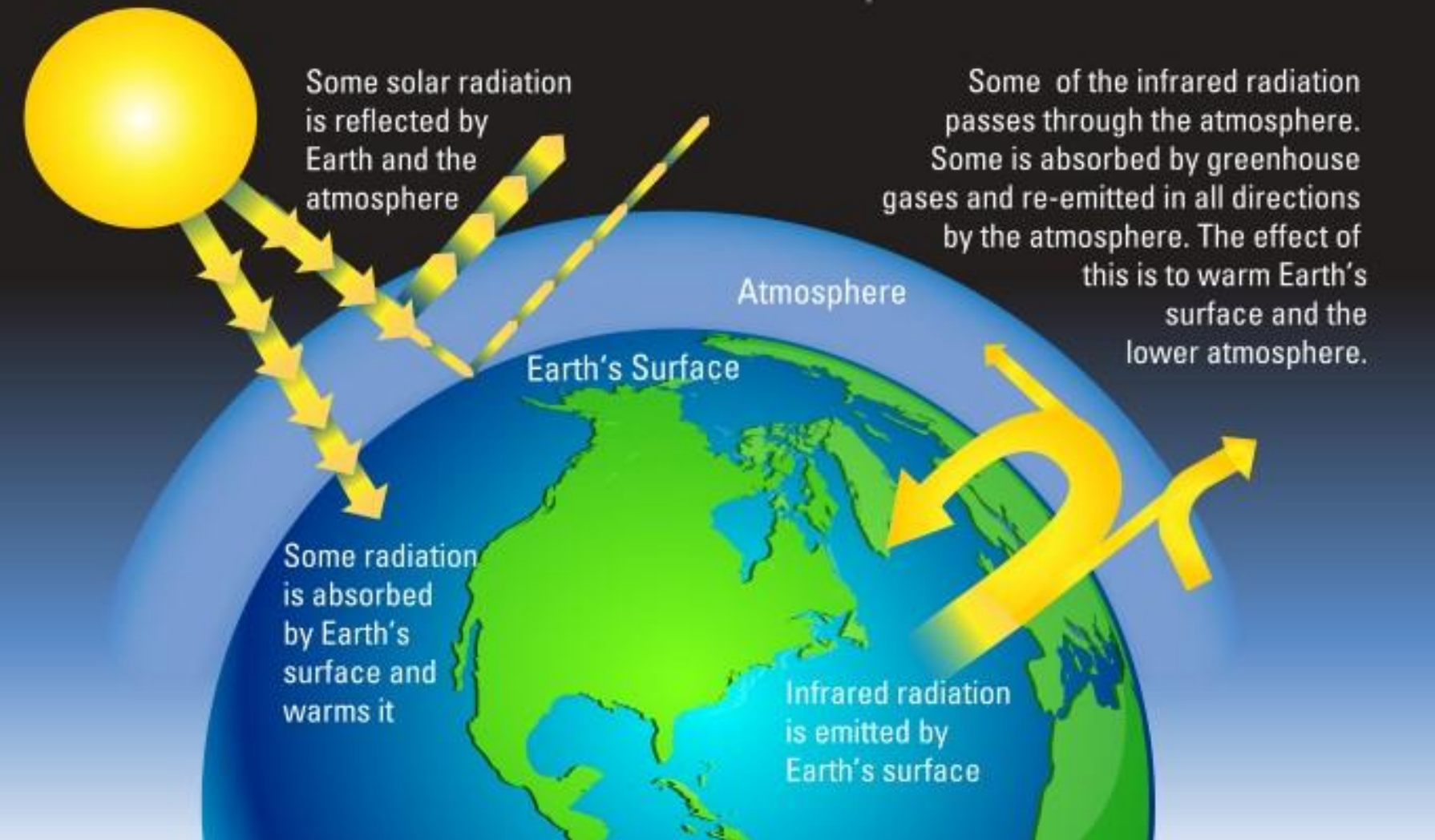
We'll come back to this.
Is the onus on us, as individuals, to change our behavior?

Or do we need systematic policies that set the right incentives?

Slightly more advanced climate science

*Disclaimer: I am not a climate or physical scientist

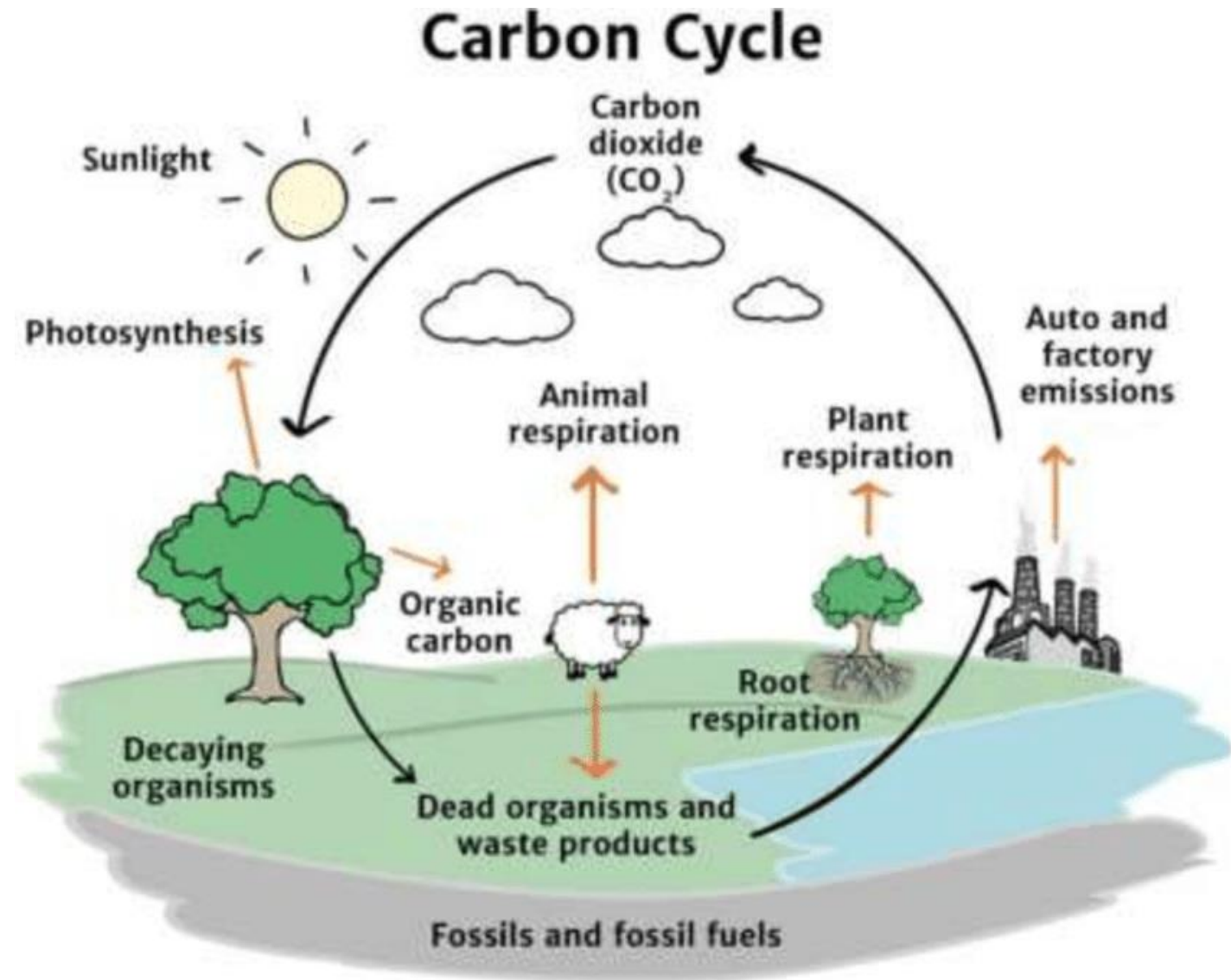
THE GREENHOUSE EFFECT



- GHGs in the atmosphere act as a “blanket” keeping the earth warmer than it otherwise would be.
 - GHGs are invisible—letting solar radiation in, without letting infrared radiation out.
- Without GHGs, global mean surface temperatures on Earth would be unlivable (-18°C or 0°F).

The carbon cycle

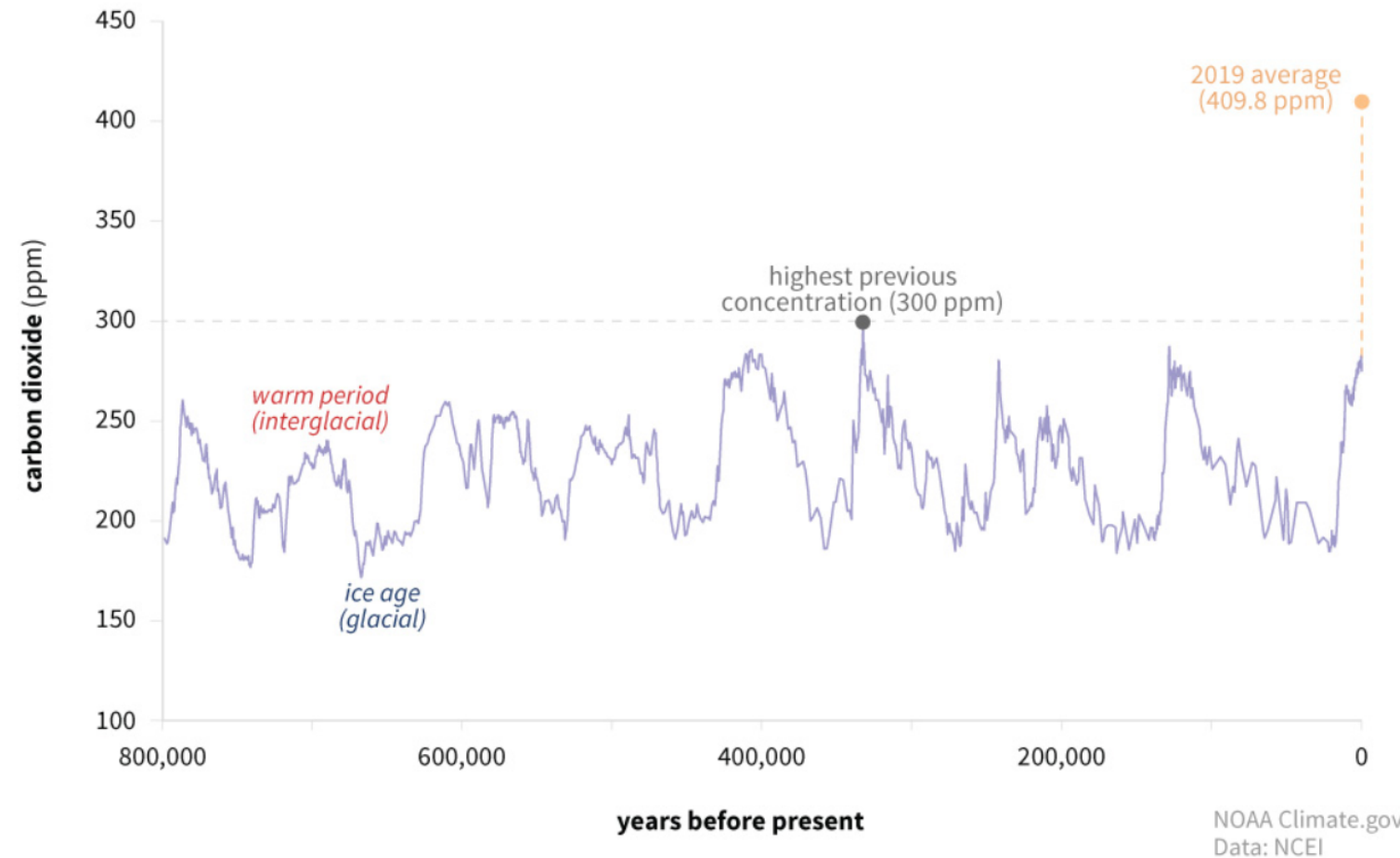
- Carbon-based molecules are the building blocks of life.
- Over millions of years, as plants and animals died and sank to the bottom of oceans and swamps, this organic matter was compressed into fossil fuels (i.e., coal, oil, natural gas).



CO₂ atmospheric concentrations

- Over the past 800,000 years, atmospheric CO₂ concentrations never exceed 300 parts-per million.
- On August 19, 2025, daily average CO₂ concentrations were 429.1 ppm
 - <https://gml.noaa.gov/ccgg/trends/monthly.html>
- Reaching ~550ppm would reflect a doubling of pre-industrial CO₂ concentrations

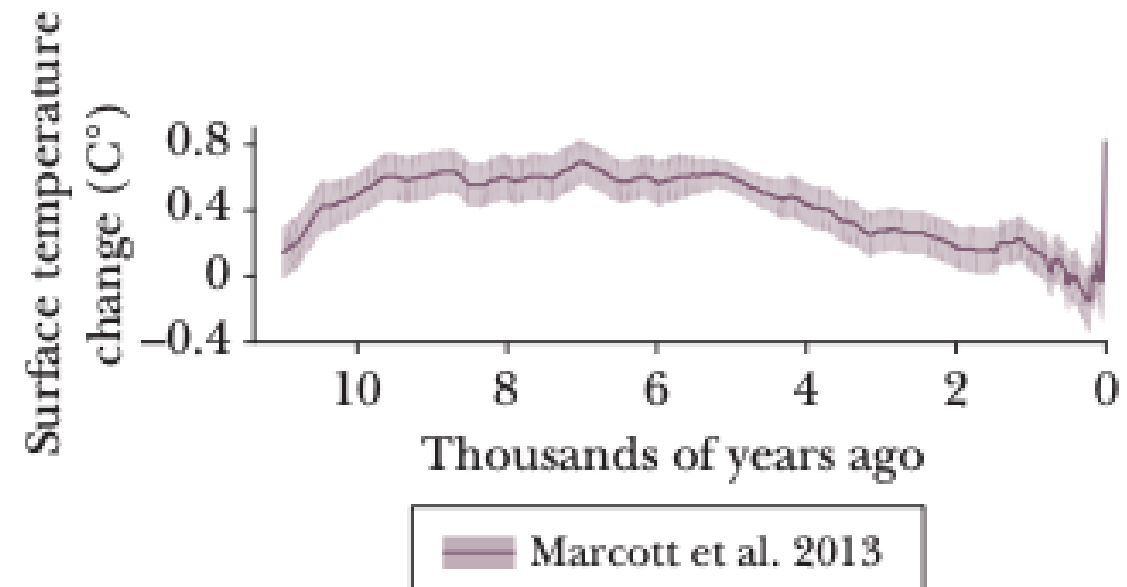
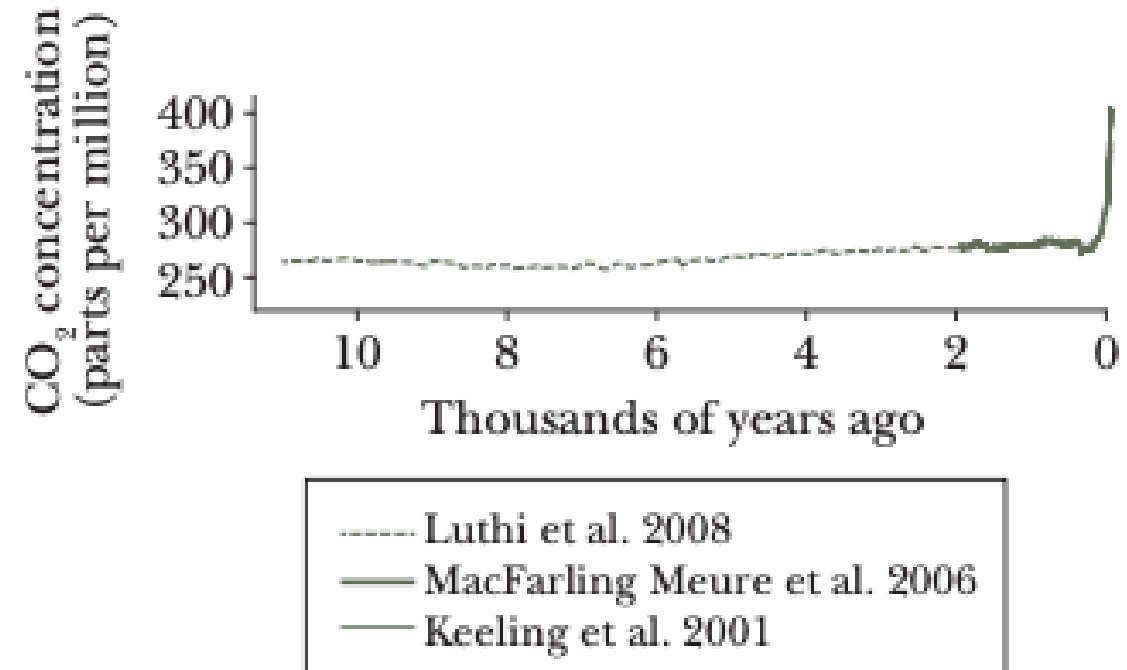
CARBON DIOXIDE OVER 800,000 YEARS



Global atmospheric carbon dioxide concentrations (CO₂) in parts per million (ppm) for the past 800,000 years. The peaks and valleys track ice ages (low CO₂) and warmer interglacials (higher CO₂). During these cycles, CO₂ was never higher than 300 ppm. On the geologic time scale, the increase (orange dashed line) looks virtually instantaneous. Graph by NOAA Climate.gov based on data from Lüthi, et al., 2008, via NOAA NCEI Paleoclimatology Program. [Update August 20, 2020. An earlier version of this image had an error in the scaling on the X axis. This affected the apparent duration and timing of the most recent ice ages, but did not affect the modern or paleoclimate carbon dioxide values.]

CO₂ atmospheric concentrations

- *Top panel:* Most recent 10,000 years, CO₂ concentrations appear flat until very recently.
- *Bottom panel:* Visual evidence of abrupt change in surface temperature corresponding to recent increase in CO₂ concentrations.



Emissions vs. concentrations

- GHG *emissions* are what's being put into the atmosphere at any given moment.
 - This is called a “**flow pollutant**”
 - *Think of water coming out of the faucet.*
- Time lag: it can take 10-50 years for CO2 emissions to increase CO2 *concentrations* in the atmosphere.
 - This is called a “**stock pollutant**”
 - *Think of the bathtub filling up.*
- CO2 concentrations dissipate slowly from the atmosphere.
 - 70% of one ton of CO2 emitted today will still be around 100 years from now.
 - *The bathtub drains much more slowly than it fills up.*
- Methane is more potent than CO2 — it's warming potential has a shorter time lag, but higher dissipation rate.

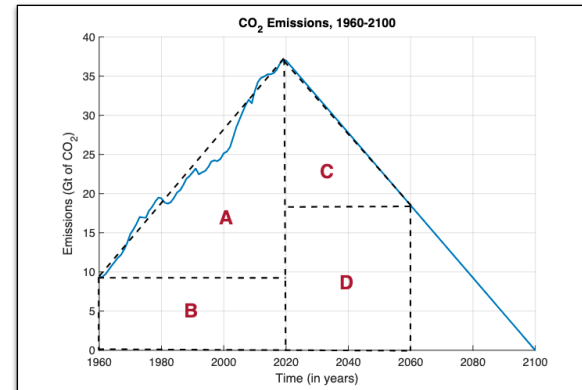


Figure 2.2: Optimistic Scenario: Rough calculation of total CO₂ emissions from 1960 through 2060. Total emissions are given by the sum of the areas of triangles A and C and rectangles B and D, which comes to 2,480 Gt. Multiplying by 0.128, this implies an increase in the atmospheric CO₂ concentration of 317 ppm.

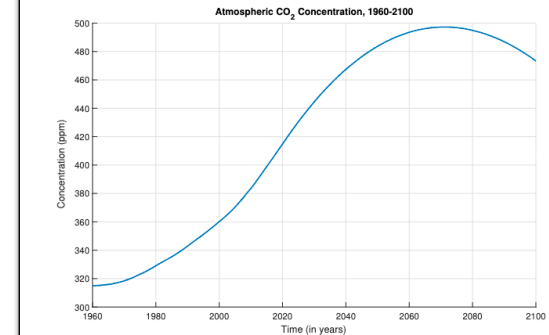


Figure 2.3: Atmospheric CO₂ concentration under optimistic scenario. By 2070 the concentration reaches a peak (of about 500 ppm) and then starts to decline because dissipation of the stock of CO₂ outweighs additions to the stock from new emissions.

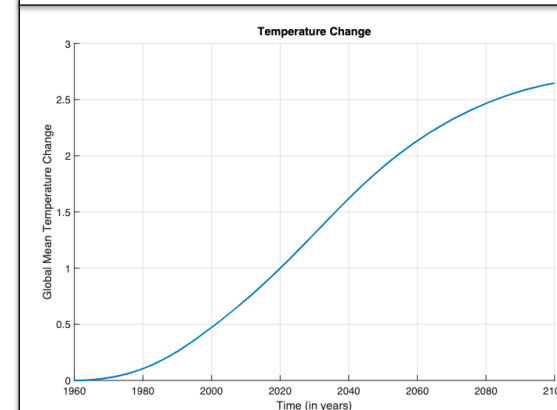


Figure 2.4: Change in temperature resulting from the change in the atmospheric CO₂ concentration under optimistic scenario. The temperature increase reaches 2°C by about 2040, and reaches about 2.7°C by the end of the century.

CO2
Emissions



CO2
Concentrations



Global
temperature
change

The **Intergovernmental Panel on Climate Change (IPCC)** is an intergovernmental body established to provide objective scientific information relevant to understanding anthropogenic climate change.

The IPCC has produced six assessment reports starting in 1990.

The sixth report (“AR6”) was released August 2021.

Scientific consensus on climate change

Table 2

Statements of the Intergovernmental Panel on Climate Change (IPCC) on Detection and Attribution of Global Climate Change

<i>First Assessment Report</i> (1990)	“Unequivocal detection of the enhanced greenhouse effect from observations is not likely for a decade or more.”
<i>Second Assessment Report</i> (1995)	“The balance of evidence suggests a discernible human influence on global climate.”
<i>Third Assessment Report</i> (2001)	“Most of the observed warming over the last 50 years is <i>likely</i> * to have been due to the increase in greenhouse gas concentration.”
<i>Fourth Assessment Report</i> (2007)	“Most of the observed increase in global average temperatures since the mid-20th century is <i>very likely</i> due to the observed increase in anthropogenic greenhouse gas concentrations.”
<i>Fifth Assessment Report</i> (2013)	“It is <i>extremely likely</i> that human influence has been the dominant cause of the observed warming since the mid-20th century.”

Source: The IPCC Assessment Reports can be found at https://www.ipcc.ch/publications_and_data/publications_and_data_reports.shtml.

* The uncertainty language used by the IPCC is precisely defined: *likely* refers to an assessed probability of at least 66 percent, *very likely* implies at least 90 percent, and *extremely likely* means at least 95 percent.

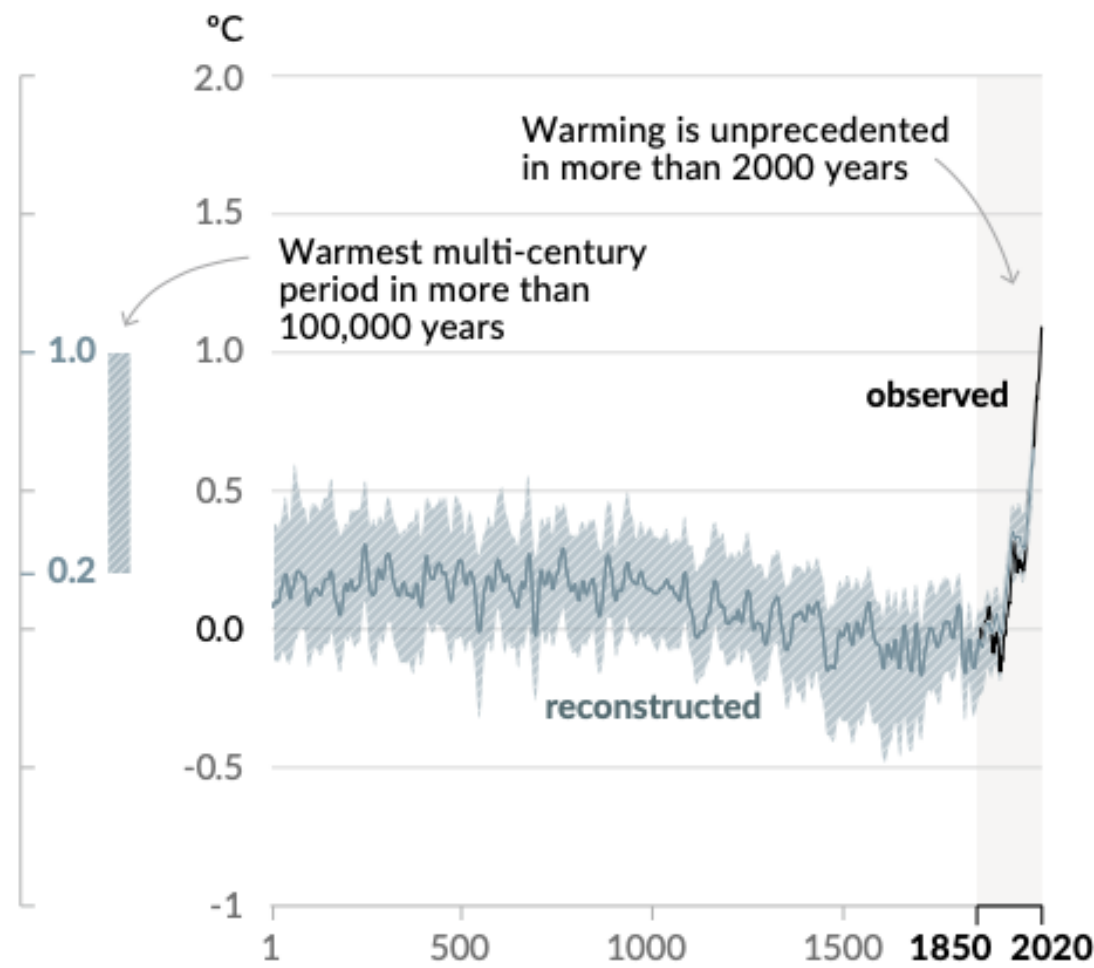
Source: Hsiang & Kopp, 2018

- Scientific consensus on anthropogenic GHG emissions and climate change.
- From Sixth Assessment Report (Aug 2021):
- **“It is unequivocal that human influence has warmed the atmosphere, ocean and land. Widespread and rapid changes in the atmosphere, ocean, cryosphere and biosphere have occurred.”**

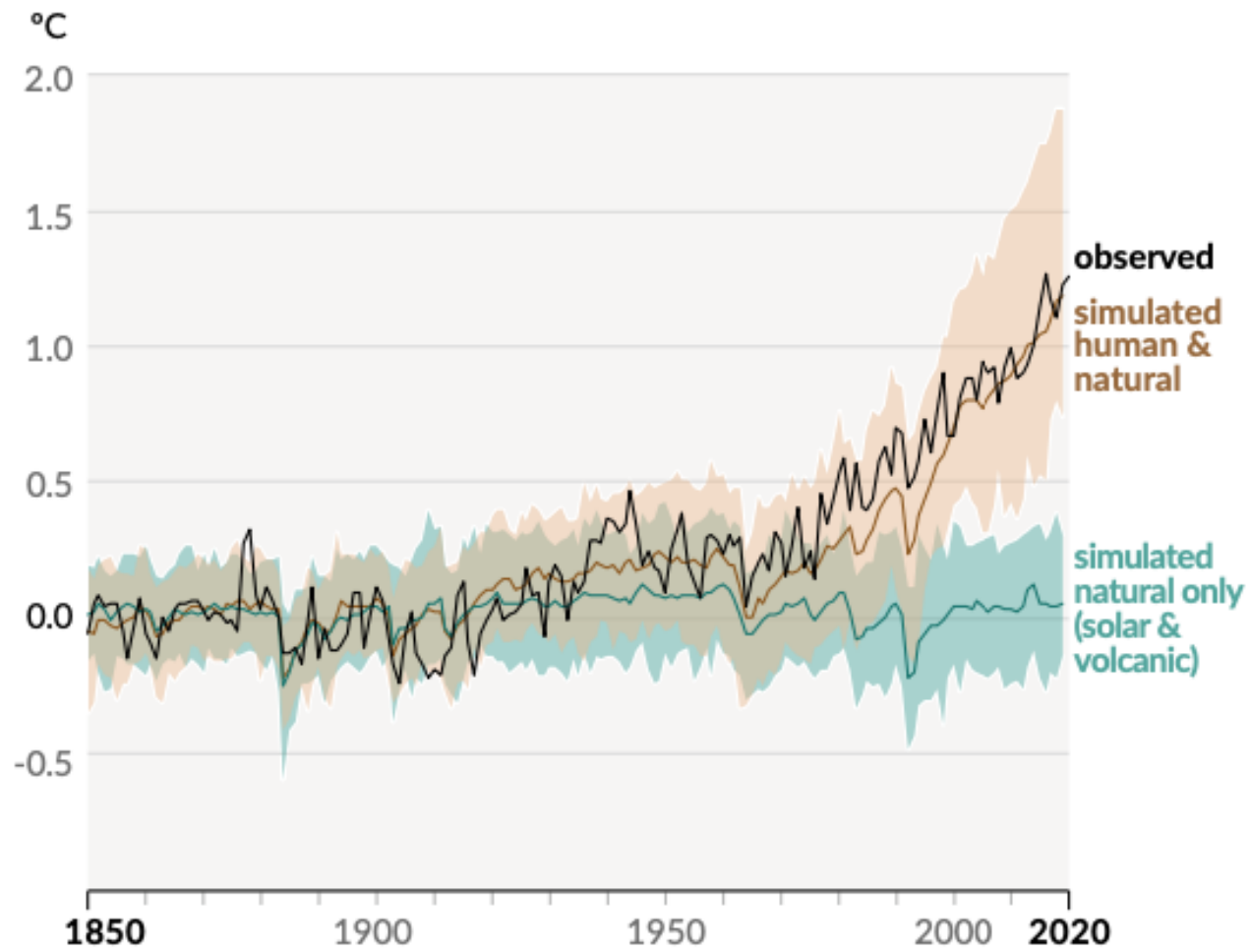
Human influence has warmed the climate at a rate that is unprecedented in at least the last 2000 years

Changes in global surface temperature relative to 1850-1900

a) Change in global surface temperature (decadal average) as **reconstructed** (1-2000) and **observed** (1850-2020)



b) Change in global surface temperature (annual average) as **observed** and simulated using **human & natural** and **only natural** factors (both 1850-2020)



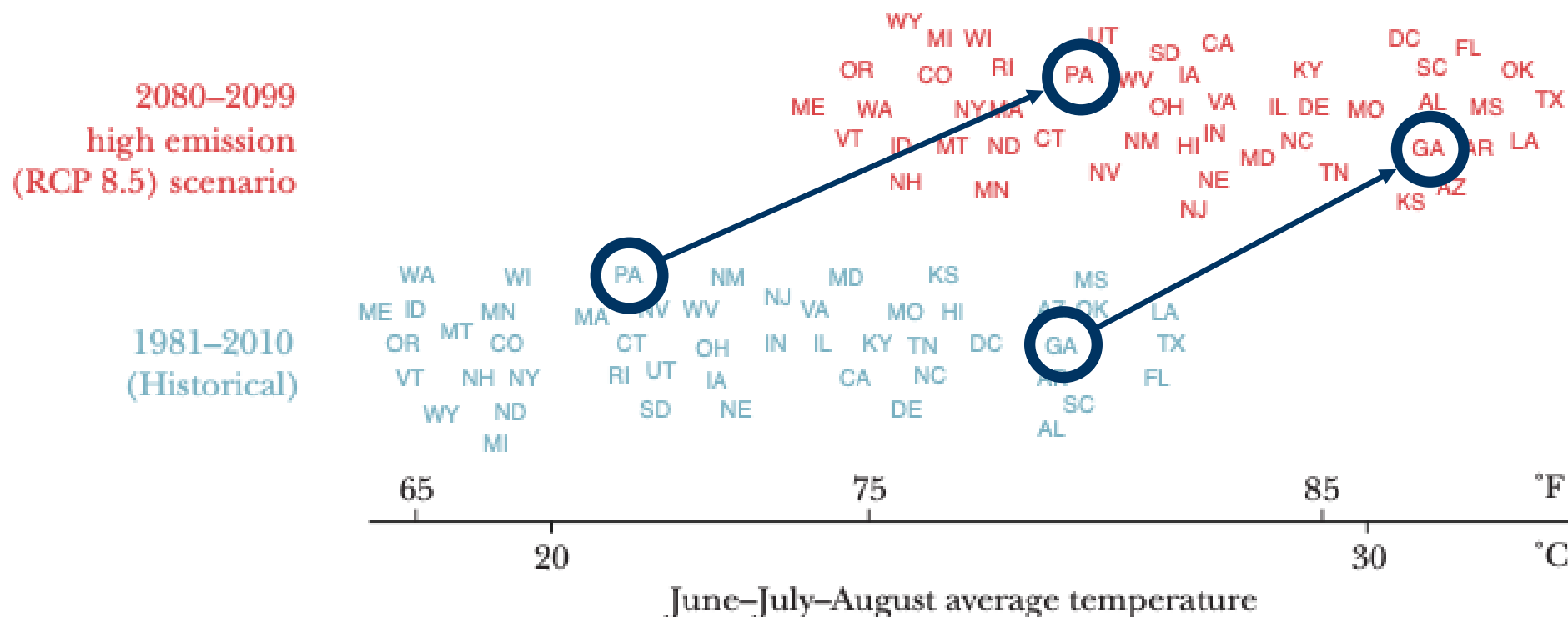
What would unmitigated climate change look like?

Figure 4

Average Temperatures for Lower 48 US States Observed during 1981–2010 and Projected for 2080–2099 in a High Emission (RCP 8.5) Scenario.

A: States (USA)

- PA summers would feel like GA summers.
- GA summers would be hotter than any current US climate.



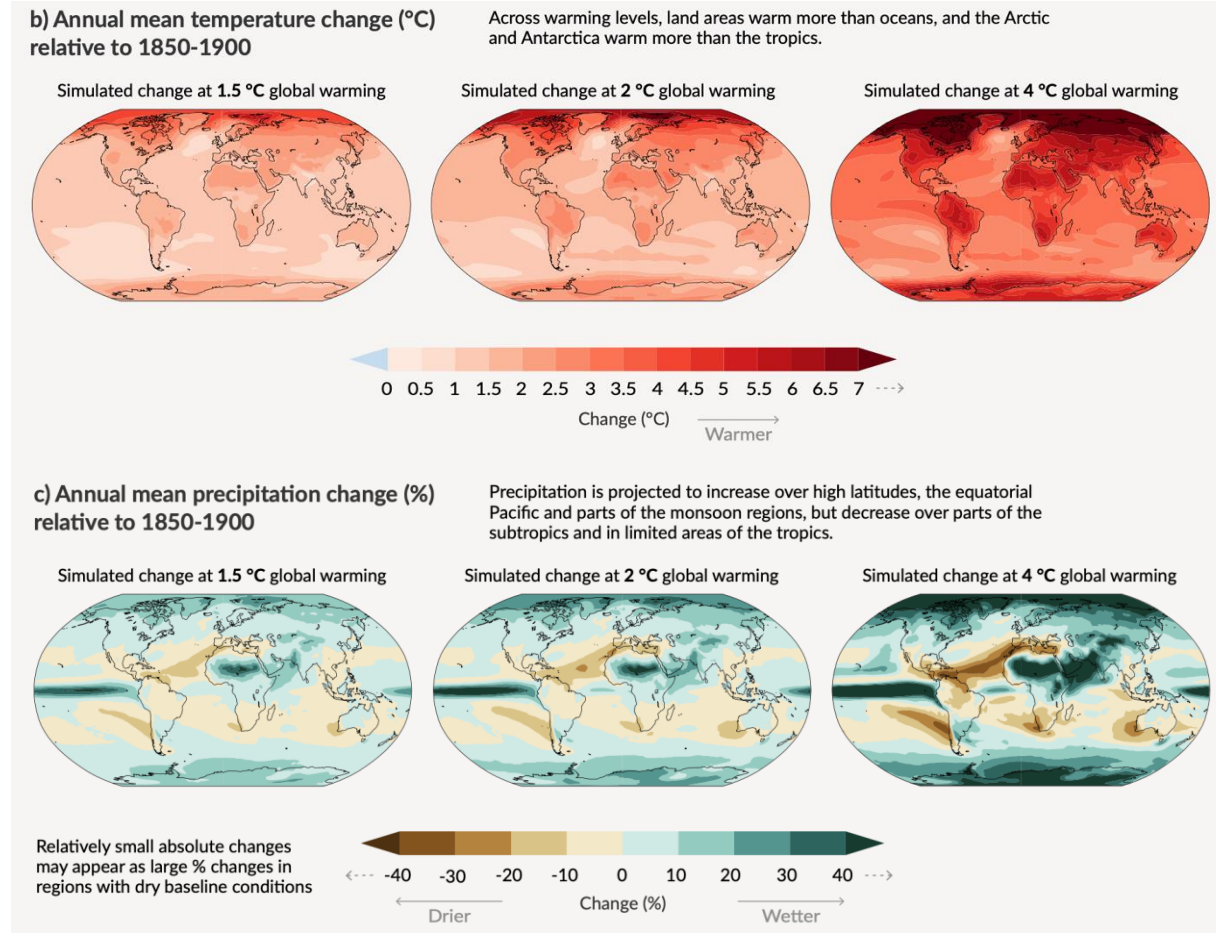
A few more (somewhat concerning) tidbits from the IPCC's AR6...

“Human-induced climate change is already affecting many weather and climate extremes in every region across the globe. Evidence of observed changes in extremes such as heatwaves, heavy precipitation, droughts, and tropical cyclones, and, in particular, their attribution to human influence, has strengthened since AR5.”

“Global surface temperature will continue to increase until at least the mid-century under all emissions scenarios considered. Global warming of 1.5°C and 2°C will be exceeded during the 21st century unless deep reductions in CO₂ and other greenhouse gas emissions occur in the coming decades.”

Many changes due to past and future greenhouse gas emissions are irreversible for centuries to millennia, especially changes in the ocean, ice sheets and global sea level.

From a physical science perspective, limiting human-induced global warming to a specific level requires limiting cumulative CO₂ emissions, reaching at least net zero CO₂ emissions, along with strong reductions in other greenhouse gas emissions.

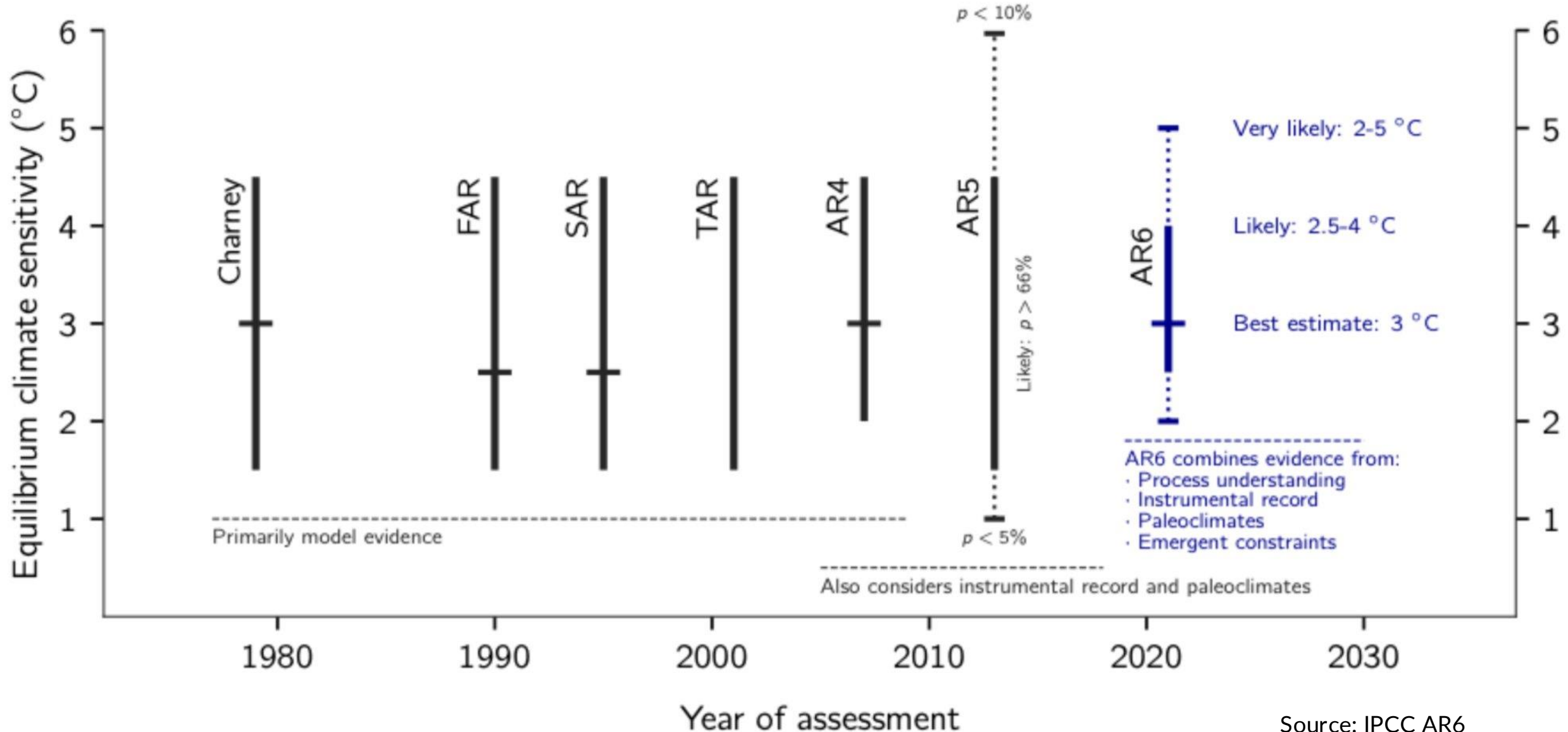


Climate sensitivity

- **Climate sensitivity** is the long-term, equilibrium temperature change in response to a doubling of atmospheric CO₂ concentrations.
 - In other words, if we double CO₂ concentrations, how much will global surface mean temperatures change?
 - This is important to understand the warming potential of anthropogenic GHG emissions.
- IPCC AR6: “...very likely range of equilibrium climate sensitivity is between 2°C (high confidence) and 5°C (medium confidence).”

Climate sensitivity

a) Evolution of equilibrium climate sensitivity assessments from Charney to AR6



What does climate sensitivity mean?

Risk scenarios

Climate scientists' best estimate is that our climate sensitivity is somewhere between 1.5 and 4.5° C. But what do those numbers mean for us? Here are some effects of climate change that the Intergovernmental Panel on Climate Change reports we will be at "high risk" of experiencing at different levels of warming.

1.5° C
(2.7° F)

All **warm-water coral reefs** shrink significantly, resulting in local extinctions.

Permafrost thaw degrades land quality in arctic regions.

The **food supply** becomes more unstable, with periodic food shocks across regions.

Wildfires become much more widespread as fire weather season grows longer.

2° C
(3.6° F)

Desertification results in shortages of water in dry regions of the world.

3° C
(5.4° F)

Ocean life suffers widespread losses in kelp and seagrass meadows and the upper ocean.

Crops yields decline in tropical regions, especially staple crops like wheat and corn.

The **food supply** suffers sustained disruptions worldwide as farmland is lost.

4.5° C
(8.1° F)

Ocean life losses become widespread in all ocean environments.

Wildfire risks are more severe, affect 100 million more people, and may be irreversible.

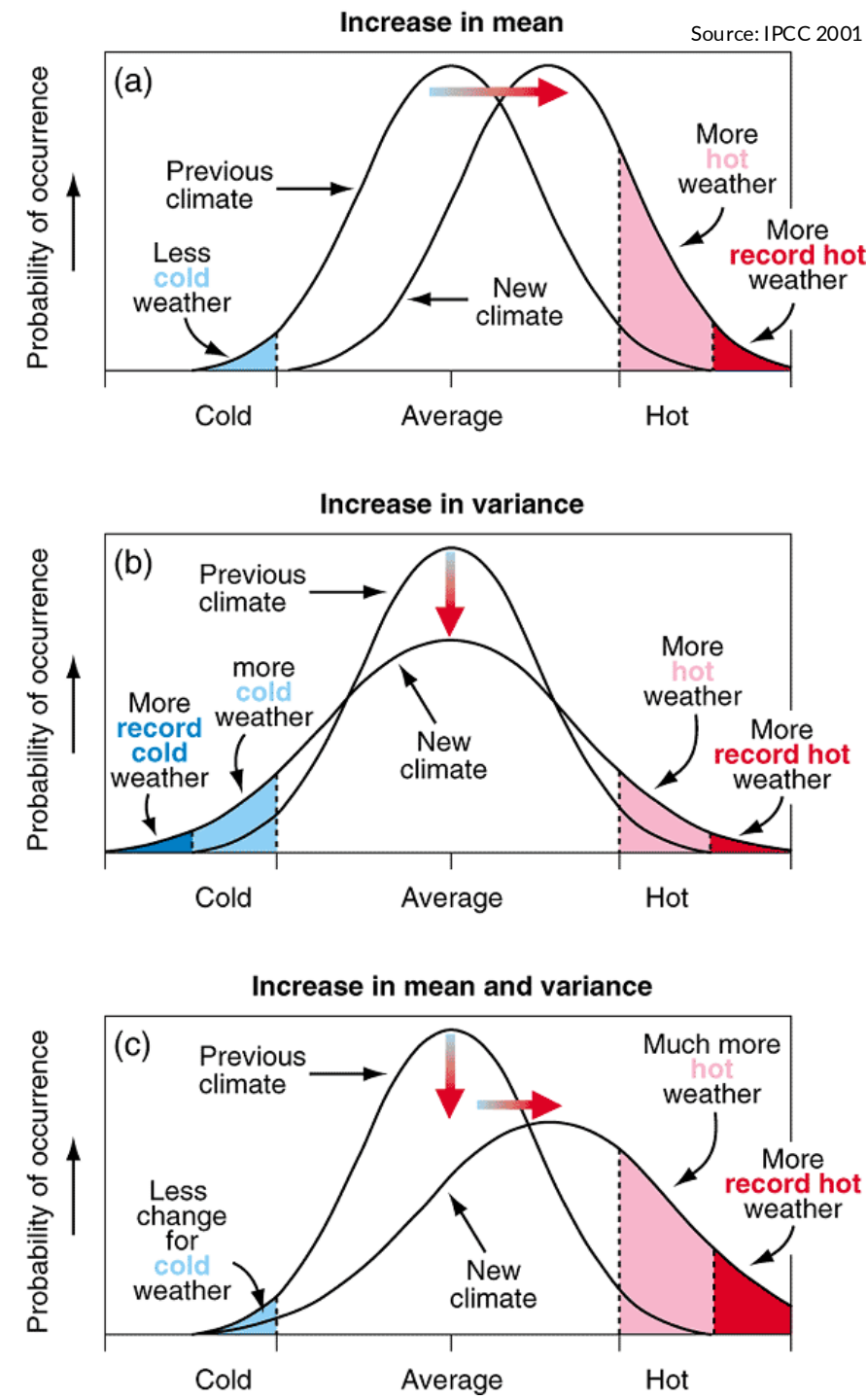
Sources: The *IPCC Special Report on Climate Change and Land*, and the *IPCC Special Report on the Ocean and Cryosphere in a Changing Climate*.

Source: <https://climate.mit.edu/explainers/climate-sensitivity>

Climate vs. weather

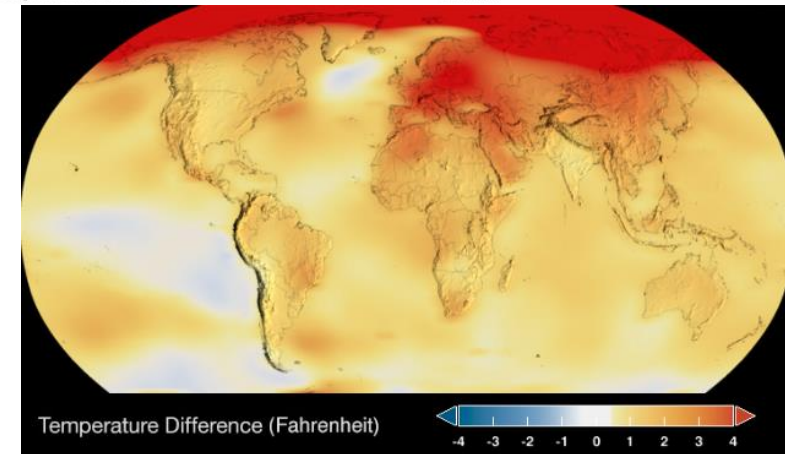
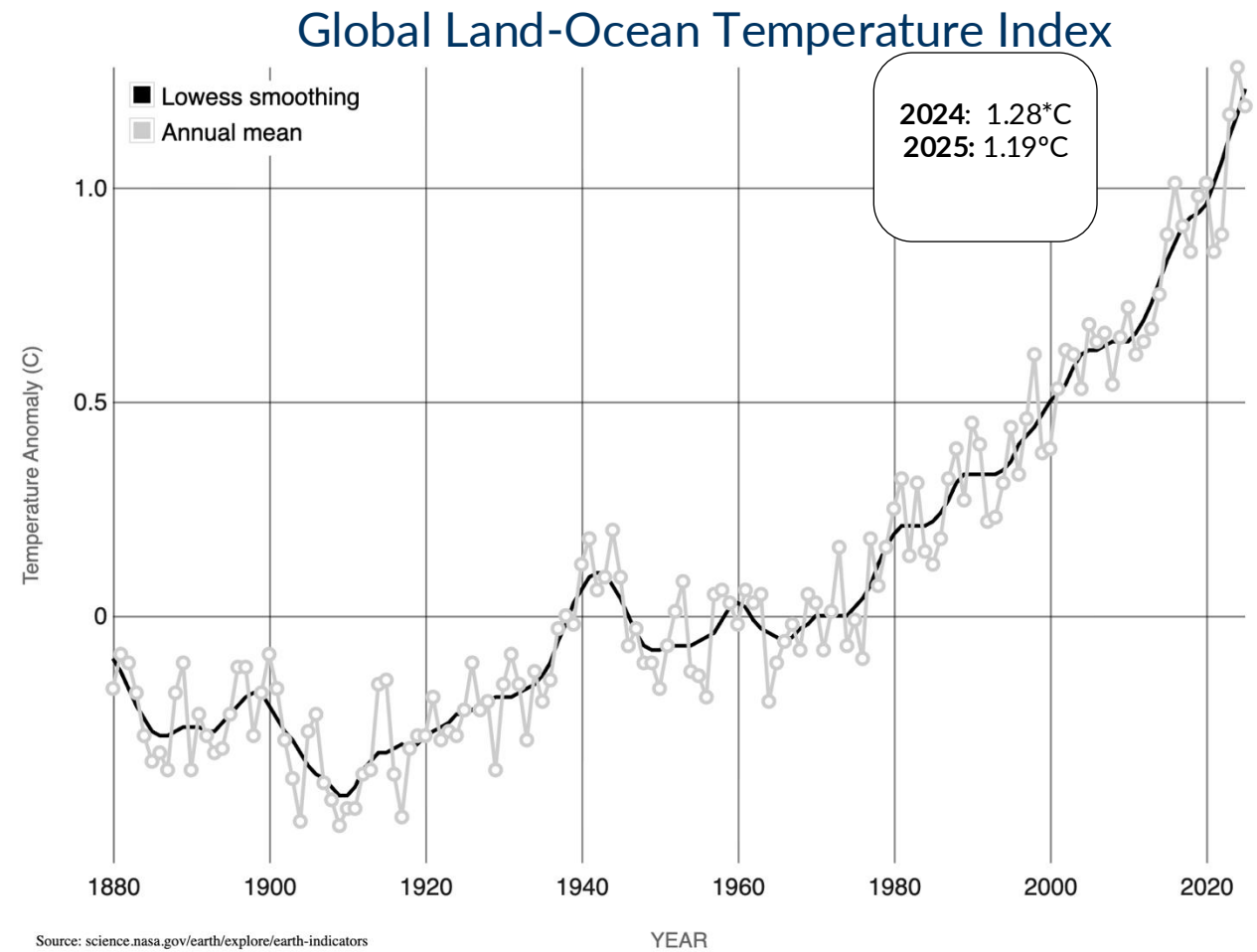
- **Climate** is the statistical average and variability of temperature, wind, humidity, cloudiness, precipitation, and other quantities over a period of time ranging from months to thousands of years.
- **Weather** is the realization of a climatic process for a short period of time.
 - I find it helpful to think of climate as a probability distribution (like a bell curve) and weather as a draw from that distribution.
- Climate *change* changes the shape, and shifts the distribution, of weather.

Nordaus (2003)



The climate is changing

- The 10 most recent years 2015–2025 are the warmest on record, ever.
- 2024 was the warmest year ever recorded (since 1880).
- Overall, Earth was about 2.65°F (1.47°C) warmer in 2024 than in the late 19th-century (1850-1900) preindustrial average.
- Visual:
<https://climate.nasa.gov/vital-signs/global-temperature/>



...but who is to blame?

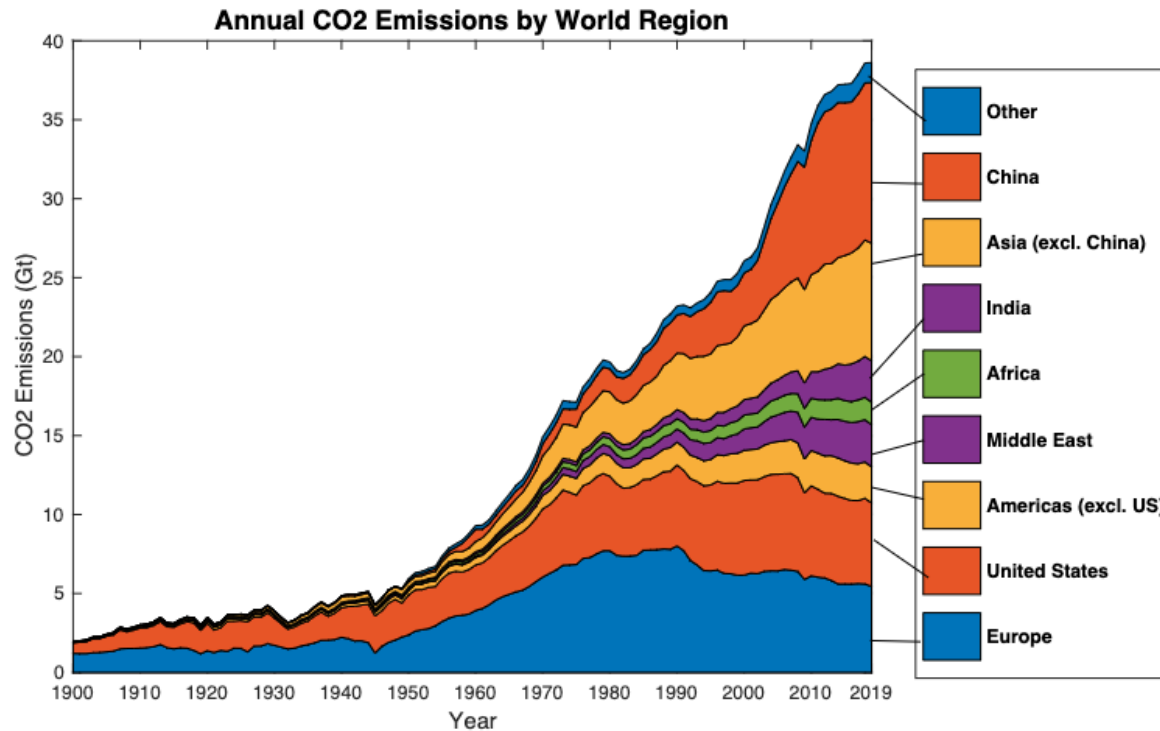


Figure 1.2: CO₂ Emissions by Region, in billions of metric tons (gigatonnes). Since 1995, emissions from North America and Europe have declined somewhat, but emissions from Asia have been rising rapidly. Source: Global Carbon Project, Supplemental Data, Global Carbon Budget, 2021.

Source: Pindyck (2021)

Table 1

Historical and Top 15 Current Emissions of Carbon Dioxide from Fossil Fuel Combustion and Cement Production

Country	Cumulative 1751–2014 (gigatonnes CO ₂)	% of Global	Emissions 2014 (gigatonnes CO ₂)	% of Global	Emissions per capita (tonnes CO ₂), 2014
China	174.7	12%	10.3	30%	7.5
United States	375.9	26%	5.3	15%	16.2
India	41.7	3%	2.2	7%	1.7
Russia / USSR	151.3	11%	1.7	5%	11.9
Japan	53.5	4%	1.2	4%	9.6
Germany	86.5	6%	0.7	2%	8.9
Iran	14.8	1%	0.6	2%	8.3
Saudi Arabia	12.0	1%	0.6	2%	19.5
South Korea	14.0	1%	0.6	2%	11.7
Canada	29.5	2%	0.5	2%	15.1
Brazil	12.9	1%	0.5	2%	2.6
South Africa	18.4	1%	0.5	1%	9.1
Mexico	17.5	1%	0.5	1%	3.8
Indonesia	11.0	1%	0.5	1%	1.8
United Kingdom	75.2	5%	0.4	1%	6.5
World	1,434.0	100%	34.1	100%	4.7

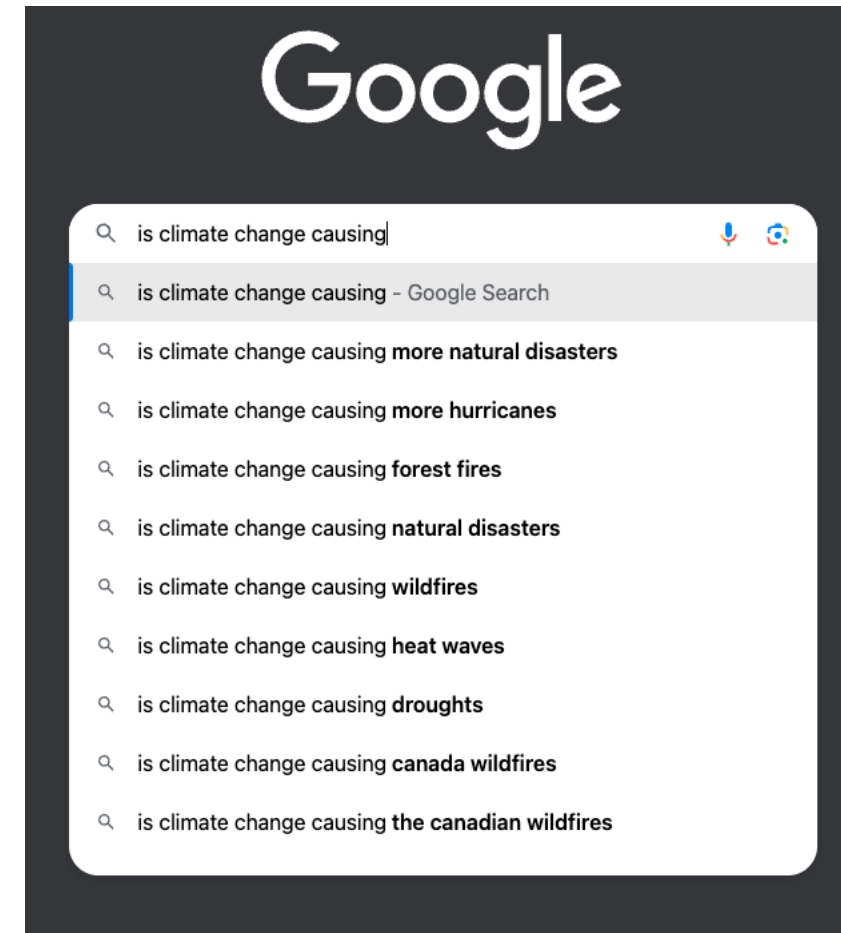
Source: Boden, Marland, and Andres (2017).

Source: Hsiang & Kopp (2018)

Climate change will change more than temperature

Climate change will lead to:

- Hotter temperatures
- Heavier and more frequent precipitation/flooding
- More frequent and severe droughts
- More intense tropical cyclones
- More (and more severe) wildfires
- Sea-level rise
- Ocean acidification (as oceans absorb more CO₂)
- Many other things (including some “unknown unknowns”)



Key takeaways about anthropogenic climate change

1. Climate change is caused primarily by burning fossil fuels (and releasing very old carbon into the atmosphere)
2. CO₂ sticks around for 300+ years. So, emitting CO₂ today affects many generations (including the current one).
3. Our climate is changing, and this affects weather and natural disasters (like forest fires, droughts, floods, and hurricanes) we are seeing today.
4. We know a lot about climate change from the physical sciences, although uncertainties about feedbacks and tipping points remain. Many additional uncertainties come from the social science side (e.g., long-run changes in population and economic growth).

**Turning our focus back to
economics....**

Economists and the Environment

Fullerton & Stavins, *Nature*, 1998

- **Myth of the universal market**
 - Market failures are environmental economists' bread and butter
- **Myth of the market solutions**
 - Market solutions can be the most cost-effective solution, but might not always work
- **Myth of market prices**
 - Environmental economists care a lot about goods and services that are not bought/sold in markets
- **Myth of efficiency**
 - Equity and distribution matters too.

For next week...

www.caseyjwichman.com/econ4210#week2

- Have a great weekend!